

Jiaxiang Liu

ARTIFICIAL INTELLIGENCE RESEARCHER · ZHEJIANG UNIVERSITY

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Summary

Jiaxiang Liu is a Postdoctoral Researcher at GDIIST, advised by Prof. Luping Shi (Tsinghua). He received his Ph.D. from Zhejiang University in 2025 under Prof. ZuoZhu Liu, and was previously a Research Intern at A*STAR Singapore with Prof. Joey Tianyi Zhou. His research focuses on **trustworthy multimodal AI for medicine**, building reliable medical vision-language models via *multi-expert chain-of-thought reasoning* and *adversarial robustness*, extending to brain decoding.

He has authored **34 peer-reviewed papers** — **10 independent first-author**, **8 co-first**, **3 corresponding**, and 13 co-author — including Cell Press *Patterns* (1, Cell Press journal, IF 10.8), *Nature Communications* (1, Nature Portfolio journal, IF 18.1), ICML (1), KDD (1), NeurIPS (1), ECCV (1), ACM MM (1), AAAI (3), EMNLP (2), MICCAI (2), ACL (1) and IEEE TCSVT (2, IF 11.1) / TMM / TIM / TETCI (5). His **Cell Press Patterns** paper presented the first clinically applicable pipeline for 3D fusion and *crown-root-bone* reconstruction, featured by Cell Press; two issued patents (IP Australia + Chinese). **H-index 16, i10-index 22** (Google Scholar).

Honours include the **National Postgraduate Scholarship**, First Prize for Outstanding Practical Achievements at Zhejiang University, Excellent Practical Achievements of Zhejiang Province, Jiangsu Jicui Postgraduate Scholarship (4 yrs), and Ni Jihong Prize; participated in **multiple NSFC and National Key R&D grants**. Reviewer for **ICML (silver, 2026)**, **NeurIPS**, **ICLR**, **KDD**, **CVPR**, **ECCV**, **MICCAI**, **AAAI**, **IEEE TCSVT/TIM**.

Education

Zhejiang University

PH.D. IN ARTIFICIAL INTELLIGENCE, COLLEGE OF COMPUTER SCIENCE AND TECHNOLOGY

- Supervisor: Prof. ZuoZhu Liu.

Hangzhou, China

2021 - 2025

Jiangxi University of Finance and Economics

M.S. IN COMPUTER TECHNOLOGY

- Supervisor: Prof. Yong Yang.

Nanchang, China

2018 - 2021

Jiangxi Normal University

B.S. IN INTERNET OF THINGS, SCHOOL OF COMPUTER INFORMATION ENGINEERING

Nanchang, China

2014 - 2018

Work Experience

Guangdong Institute of Intelligence Science and Technology (GDIIST)

POSTDOCTORAL RESEARCHER

- Supervisor: Prof. Luping Shi (Tsinghua University).

Hengqin, Zhuhai, China

2025 - PRESENT

Agency for Science, Technology and Research (A*STAR)

RESEARCH INTERN

- Supervisor: Prof. Joey Tianyi Zhou.

Singapore

Sep. 2023 - Sep. 2024

Publications

First Author (Independent) × 10

(1 Cell Press journal, 6 CCF-A / TH-CPL-A conferences, 2 JCR Q1 journals, 1 flagship medical conference)

Co-First Author × 8 | **Corresponding Author** × 3 | **Co-Author** × 13 (incl. 1 Nature Communications)

Symbols: # co-first author, * corresponding author.

SELECTED FIRST-AUTHOR CONFERENCE PAPERS

BioMedVR: Confusion-Aware Mixture-of-Prompt Experts for Biomedical Visual Reprogramming [PDF]

ECCV 2026

JIAXIANG LIU, TIANXIANG HU, JUWEI GUAN, YUJIE WU, YUSONG WANG, YAO MU, ZUOZHU LIU, MINGKUN XU

2026

CCF-B / TH-CPL-A. First input-space dual-expert visual reprogramming for biomedical VLMs.

Self-Calibrated Consistency can Fight Back for Adversarial Robustness in Vision-Language Models [PDF] JIAXIANG LIU , JIAWEI DU, XIAO LIU, SHANGYANG LI, SONGCHEN MA, CHANGSHUO WANG, PRAYAG TIWARI, MINGKUN XU CCF-A / TH-CPL-A. arXiv:2510.22785; code at github.com/JXLiu-AI/SCC	ICML 2026 2026
MindAdapter: Few-Shot Parameter-Efficient Residual Calibration of Cross-Subject Brain-to-Visual Decoding Models [PDF] JIAXIANG LIU , JIAWEI DU, XUPENG CHEN, GUOQI LI, JIANG CAI, SIMON FONG, MINGKUN XU CCF-A / TH-CPL-A. Project page: jxliu-ai.github.io/MindAdapter	KDD 2026 (AI4Sciences track) 2026
KPL: Training-Free Medical Knowledge Mining of Vision-Language Models [PDF] JIAXIANG LIU , TIANXIANG HU, JIAWEI DU, RUIYUAN ZHANG, JOEY TIANYI ZHOU, ZUOZHU LIU CCF-A. Code: github.com/JXLiu-AI/KPL	AAAI 2025 (Oral) 2025
VPL: Visual Proxy Learning Framework for Zero-Shot Medical Image Diagnosis [PDF] JIAXIANG LIU , TIANXIANG HU, HUIMIN XIONG, JIAWEI DU, YANG FENG, JIAN WU, JOEY TIANYI ZHOU, ZUOZHU LIU TH-CPL / CAAL A	EMNLP 2024 2024
MedCoT: Medical Chain of Thought via Hierarchical Expert [PDF] JIAXIANG LIU , YUAN WANG, JIAWEI DU, JOEY TIANYI ZHOU, ZUOZHU LIU TH-CPL / CAAL A. Code: github.com/JXLiu-AI/MedCoT	EMNLP 2024 (Oral) 2024
ToothSegNet: Image Degradation Meets Tooth Segmentation in CBCT Images [PDF] JIAXIANG LIU , TIANXIANG HU, YANG FENG, WANGHUI DING, ZUOZHU LIU Flagship medical conference. Code: github.com/JXLiu-AI/Tooth-UNet	IEEE ISBI 2022
FIRST-AUTHOR JOURNAL PAPERS	
Deep learning-enabled 3D multimodal fusion of cone-beam CT and intraoral mesh scans for clinically applicable tooth-bone reconstruction [PDF] JIAXIANG LIU , JIN HAO, HANGZHENG LIN, WEI PAN, JIANFEI YANG, YANG FENG, GAOANG WANG, JIN LI, ZHIHE ZHAO, ZUOZHU LIU Cell Press data-science journal	Cell Press Patterns 2023
Parameter-Efficient Transfer Learning for Medical Visual Question Answering [PDF] JIAXIANG LIU , TIANXIANG HU, YAN ZHANG, YANG FENG, JIN HAO, JUNHUI LV, ZUOZHU LIU JCR Q1.	IEEE TETCI 2023
IFCoT: Insights Fusion Chain of Thought for Multimodal Reasoning in LLMs JIAXIANG LIU , JIAWEI DU, TIANXIANG HU, LINJIE TONG, YAN ZHANG, BING LI, YANG FENG, JIAN WU, JOEY TIANYI ZHOU, ZUOZHU LIU JCR Q1 / SCI Q1. IF 11.1	IEEE TCSVT (Accepted) 2026
Modest-Align: Data-Efficient Alignment for Vision-Language Models [PDF] JIAXIANG LIU , YUAN WANG, JIAWEI DU, JOEY TIANYI ZHOU, MINGKUN XU, ZUOZHU LIU JCR Q1. arXiv:2510.21606; code: github.com/JXLiu-AI/Modest-Align	Pattern Recognition (Under Review) 2025
CO-FIRST AUTHOR / CORRESPONDING AUTHOR PAPERS (SELECTED)	
3D-RAD: A Comprehensive 3D Radiology Med-VQA Dataset with Multi-Temporal Analysis and Diverse Diagnostic Tasks [PDF] XIAOTANG GAI, JIAXIANG LIU #, YU LI, ZIJIE MENG, JIAN WU, ZUOZHU LIU CCF-A / TH-CPL-A. Co-first author	NeurIPS 2026 (Dataset Track) 2026
Med-GLIP: Advancing Medical Language-Image Pre-training with Large-scale Grounded Dataset [PDF] ZIYE DENG, RUIHAN HE, JIAXIANG LIU #, YUAN WANG, ZIJIE MENG, SONGTAO JIANG, YONG XIE, ZUOZHU LIU CCF-B. Co-first author	IEEE BIBM 2025
GTCL: Graph-Text Contrastive Learning Meets Log Anomaly Detection [PDF] XIAO TAN, JIAXIANG LIU #, YUJIE WU, MINGKUN XU CCF-B. Co-first author	IEEE ICASSP 2026
Fair-MoE: Fairness-Oriented Mixture of Experts in Vision-Language [PDF] LINJIE TONG, PEIRAN WANG, JIAXIANG LIU *, ZUOZHU LIU CCF-B. Corresponding author	MICCAI 2025

Infrared and Visible Image Fusion Based on Modal Feature Fusion Network and Dual Vision Salient Decision YONG YANG, JIAXIANG LIU* , SHUYING HUANG, WEIGUO WAN, XIANGKAI KONG, WANG ZHANG CCF-B. Corresponding author.	IEEE ICME (Oral) 2021
V2T-CoT: From Vision to Text Chain-of-Thought for Medical Reasoning and Diagnosis [PDF] YUAN WANG, JIAXIANG LIU# , ZUOZHU LIU CCF-B. Co-first author	MICCAI 2025
Scalable Geometric Fracture Assembly via Co-creation Space among Assemblers [PDF] RUIYUAN ZHANG, JIAXIANG LIU# , ZEXI LI, DONG HAO, JIE FU, CAO WU CCF-A. Co-first author	AAAI 2024
PVP: Polar Representation Boost for 3D Semantic Occupancy Prediction YUJING XUE, JIAXIANG LIU# , JIAWEI DU, JOEY TIANYI ZHOU CAAI B. Co-first author	WACV 2025
MedThink: Explaining Medical Visual Question Answering via Multimodal Decision-Making Rationale XIAOTANG GAI, CHENYI ZHOU, JIAXIANG LIU# , YANG FENG, JIAN WU, ZUOZHU LIU CCF-B. Co-first author	NAACL Findings 2025
TSNet: Integrating Dental Position Prior and Symptoms for Tooth Segmentation from CBCT Images LINJIE TONG, JIAXIANG LIU# , YANG FENG, ZUOZHU LIU Co-first author.	MIDL 2023
Co-AUTHOR PAPERS (SELECTED)	
DentalDiff: Angle-Controllable CBCT Projection Synthesis from a Single Panoramic X-ray WEN MA, ZIKAI XIAO, JIAXIANG LIU , YANG FENG, ZUOZHU LIU CCF-A. Co-author.	ACM Multimedia 2026 (Main Track) 2026
DentVLM: A Multimodal Vision-Language Model for Comprehensive Dental Diagnosis and Enhanced Clinical Practice [PDF] ZIJIE MENG, JIN HAO, XIWEI DAI, YANG FENG, JIAXIANG LIU , ET AL., ZUOZHU LIU IF 15.7. arXiv:2509.23344	Nature Communications (Accepted) 2026
BMAM: Brain-inspired Multi-Agent Memory Framework [PDF] YANG LI, JIAXIANG LIU , YUSONG WANG, YUJIE WU, MINGKUN XU CCF-A. arXiv:2601.20465	ACL 2026 2026
Beyond N-grams: A Hierarchical Reward Learning Framework for Clinically-Aware Medical Report Generation [PDF] YUAN WANG, SHUJIAN GAO, JIAXIANG LIU , SONGTAO JIANG, HAOXIANG XIA, XIAOTIAN ZHANG, ZHAOLU KANG, YEMIN WANG, ZUOZHU LIU CCF-A.	AAAI 2026 2026
HiCT: High-Precision 3D CBCT Reconstruction From A Single X-ray WEN MA, JIAXIANG LIU , ZIKAI XIAO, ZIYANG WANG, YANG FENG, ZUOZHU LIU CCF-B	IEEE ICASSP 2026
Infrared and Visible Image Fusion via Texture Conditional Generative Adversarial Network YONG YANG, JIAXIANG LIU , SHUYING HUANG, WEIGUO WAN, WENYING WEN, JUWEI GUAN JCR Q1 TOP. Code: github.com/JXLiu-AI/TC-GAN	IEEE TCSVT 2021
End-to-End Rain Removal Network Based on Progressive Residual Detail Supplement YONG YANG, JUWEI GUAN, SHUYING HUANG, WEIGUO WAN, YATING XU, JIAXIANG LIU JCR Q1 TOP.	IEEE TMM 2021
Infrared and Visible Image Fusion based on Dual-kernel Side Window Filtering and S-shaped Curve Transformation YONG YANG, WANG ZHANG, SHUYING HUANG, WEIGUO WAN, JIAXIANG LIU , XIANGKAI KONG JCR Q1.	IEEE TIM 2021

ECLB: Efficient Contrastive Learning on Bi-Level for Noisy Labels

JUWEI GUAN, **JIAXIANG LIU**, SHUYING HUANG, YONG YANG

JCR Q1 / SCI Q1.

Knowledge-Based Systems (KBS)

2024

A ChatGPT-Aided Explainable Framework for Zero-Shot Medical Image Diagnosis [PDF]

JIAXIANG LIU, TIANXIANG HU, YAN ZHANG, XIAOTANG GAI, YANG FENG, ZUOZHU LIU

ICML-IMLH Workshop

2023

Patents

Multiple Decision Maps Based Infrared and Visible Image Fusion

YONG YANG (SUPERVISOR); **LIU, JIAXIANG**; ET AL.

IP Australia, No. 2020100178

2021

Image Fusion Method, Device, Equipment and Readable Storage Medium

YANG FENG; ZUOZHU LIU; **LIU, JIAXIANG**

Chinese Patent, No. 202311022450

2023

Research Projects

Multi-Expert Verified Medical Chain-of-Thought Reasoning (MedCoT framework)

SCIENTIFIC RESEARCH FUND OF ZHEJIANG UNIVERSITY

Hangzhou, China

2023 - 2025

- Hierarchical experts + Sparse MoE + feature-level voting; a 256M model outperforms LLaVA-Med (7B) by **5%+**.
- **Mentoring**: mentored 2 master students, who published 4 papers at **NeurIPS, NAACL, AAAI, MICCAI**.
- **Personal outputs**: 4 top-tier conference papers as lead/co-lead author (**MedCoT, KPL, VPL, MedThink**); awarded **First Prize** at Zhejiang University and the **Outstanding Practical Achievement Award** of Zhejiang Province.

Clinical applicable multimodal fusion algorithm research for 3D intraoral scanning and CBCT data

SCIENTIFIC RESEARCH FUND OF ZHEJIANG UNIVERSITY

Hangzhou, China

2022 - 2023

- Reduced multimodal oral data *crown-root-bone* reconstruction time from a conventional **5-7 hours** to **20-25 minutes** (~15× speed-up).
- Algorithm deployed in the **Angelalign IRS** clinical orthodontic system; results featured by **Cell Press** in an official news release.
- **Mentoring**: mentored 1 undergraduate student, who published 3 papers at **MIDL, MICCAI, BIBM**.
- **Personal outputs**: **Cell Press Patterns 2023** first-author paper; awarded **First Prize** at Zhejiang University and **Excellent Practical Achievement** of Zhejiang Province.

Infrared and visible image fusion for public safety

STUDENT RESEARCH PROJECT OF JIANGXI UNIVERSITY OF FINANCE AND ECONOMICS

Nanchang, China

2020 - 2021

- Designed multi-scale and modal feature fusion networks for IR-VIS image fusion in public-safety scenarios.
- Outputs: 4 IEEE journal / conference papers (TCSVT, TMM, TIM, ICME Oral) and 1 international patent (IP Australia).

Skills & Service

Programming Python, MATLAB, C, HTML5, SQL.

Languages Native Chinese; Fluent English.

Editorial service Young Editorial Board Member, *Computer and Modernization*.

Conference reviewer ICML (silver tier, 2026), NeurIPS, ICLR, KDD, CVPR, ECCV, MICCAI, AAAI.

Journal reviewer IEEE TCSVT, IEEE TIM, IEEE TIP.

Honors & Awards

2024	First Prize (university level) , Outstanding Practical Achievements of Professional Degree Graduate Students, Zhejiang University	Hangzhou, China
2024	Provincial Award , Outstanding Practical Achievements of Professional Degree Graduate Students, Zhejiang Province	Zhejiang, China
2022–2025	Multiple Graduate Scholarships , Jiangsu Province Jicui Postgraduate Scholarship (4 yrs) & Ni Jihong Prize, Zhejiang University	China
2021	National Scholarship , National Scholarship for Postgraduates	China
2019–2020	National & Provincial 1st–3rd Prizes (5 total) , Huawei Cup China Post-Graduate Mathematical Modeling (16th 2nd; 17th 3rd) & Jiangxi Province Modeling (1st, 2nd) & 7th TipDM Cup Data Mining (2nd)	China
2018–2025	Excellent Graduate , Jiangxi Normal University & Zhejiang University	China
2015	Provincial First Prize , Jiangxi Province University Student Science and Technology Innovation and Skills Competition	Jiangxi, China